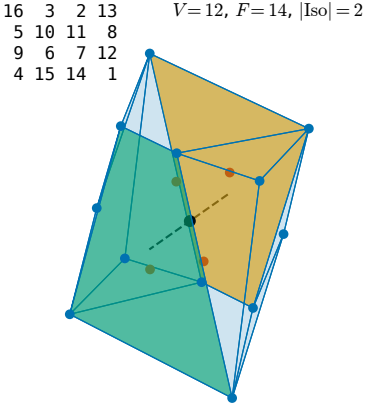
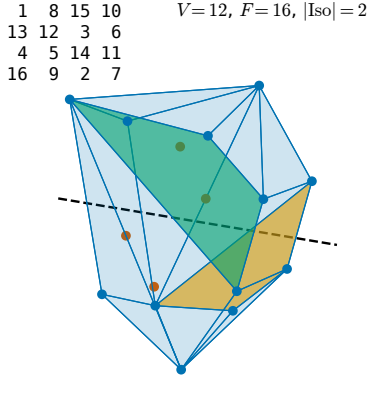
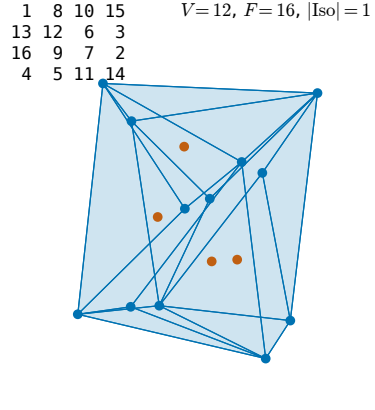


(a) centrally symmetric
(isometry $\mathbf{p} \mapsto -\mathbf{p}$)



(b) self-complementary
(half-turn about the dashed axis)



(c) generic
(no non-trivial isometry)